Question	Answer	Marks	Partial Marks
9(d)	1.5 oe	**2**	**M1** for $4^{2x-1} = 4^2$ or better
9(e)	$a = 3$ $b = 4$ $c = –59$ $d = –20$	**3**	**B2** for 3 correct values or for correct unsimplified expanded expression or for simplified four-term expression of correct form with 3 terms correct or **B1** for 2 correct values or for correct expansion of one pair of brackets with at least 3 out of 4 terms correct.
10(a)(i)	(15, 6)	**2**	**B1** for each
10(a)(ii)	$\begin{pmatrix} 3 \\ 24 \end{pmatrix}$	**1**	
10(a)(iii)	13.6 or 13.60…	**2**	**M1** for $(–11)^2 + 8^2$ oe
10(b)(i)	$\mathbf{a} + \frac{3}{5}(\mathbf{b} – \mathbf{a})$ or $\mathbf{b} + \frac{2}{5}(\mathbf{a} – \mathbf{b})$ leading to $\frac{2}{5}\mathbf{a} + \frac{3}{5}\mathbf{b}$ with no errors	**M3**	**M2** for [$\overrightarrow{MR}$ =] $\frac{3}{5}(\mathbf{b} – \mathbf{a})$ oe or [$\overrightarrow{NR}$ =] $\frac{2}{5}(\mathbf{a} – \mathbf{b})$ oe or **M1** for $\overrightarrow{MN} = \mathbf{b} – \mathbf{a}$ or $\overrightarrow{NM} = \mathbf{a} – \mathbf{b}$ or a correct route for $\overrightarrow{OR}$
10(b)(ii)(a)	$k = 5$, $c = 10$	**4**	**B2** for $c = 10$ or **M1** for $c(\frac{2}{5}\mathbf{a} + \frac{3}{5}\mathbf{b}) = \mathbf{b} + 4\mathbf{a} + k\mathbf{b}$ oe or for $\frac{2}{5}c = 4$ and **M1** for $\frac{3}{5} \times$ *their* $c = k + 1$
10(b)(ii)(b)	$3\mathbf{a} + 6\mathbf{b}$ final answer	**1**	**FT** $3\boldsymbol{a} + (\textit{their } k + 1)\boldsymbol{b}$
11(a)	$3x^2 – 8x – 3$	**2**	**B1** for two terms correct or correct answer seen

Question	Answer	Marks	Partial Marks
11(b)	$3x^2 - 8x - 3 = 0$	**M1**	**FT** their **part (a)**
	Correct method to solve their 3-term quadratic $(3x + 1)(x - 3)$ [=0] $\frac{-(-8) \pm \sqrt{(-8)^2 - 4(3)(-3)}}{2(3)}$	**M2**	**M1** for $(3x + a)(x + b)$ [=0] where $ab = -3$ or $3b + a = -8$ or for $\sqrt{(-8)^2 - 4(3)(-3)}$ or for $\frac{p \pm \sqrt{q}}{r}$ where $p = -(-8)$ and $r = 2(3)$ seen or for a correct method for solving a 2-term quadratic
	(3, –18) $\left(-\frac{1}{3}, \frac{14}{27}\right)$	**B2**	**B1** for one correct point or for two correct x-values, or **M1** for substitution of *their* x-values into $y = x^3 - 4x^2 - 3x$ **shown**

Question	Answer	Marks	Partial Marks
11(c)	(3, –18) minimum with reason $\left(-\frac{1}{3}, \frac{14}{27}\right)$ maximum with reason	**3**	Reasons could be e.g. 1. A reasonable sketch of a positive cubic 2. Correct use of 2nd derivative = $6x - 8 = 10$, $10 > 0$, so (3, –18) is a minimum oe. 2nd derivative = $6x - 8 = -10$, $-10 < 0$ so $\left(-\frac{1}{3}, \frac{14}{27}\right)$ is a maximum oe. 3. Evaluates correctly values of y on both sides of both correct stationary points 4. Finds gradient on each side of both correct stationary points. **B2** for 1 correct with a reason for that stationary point or for **both** x-values correct with correct conclusions and reasonable sketch of a positive cubic, or for correct substitution of **both** of *their* x-values into *their* second derivative shown, or substitution shown for one x-value either side of **both** of *their* stationary points to find the gradients. Or **M1** for showing [2nd derivative =] $6x - 8$ or substitution shown for one x-value either side of **one** of *their* stationary points to find the gradients. or for reasonable sketch of positive cubic.